

q1_analysis

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Contents

sensitivity analysis (PMM)	35
--------------------------------------	----

Import and clean dataset

```

pancreatic_df <- read_csv("data/eb1.csv") |>
  janitor::clean_names() |>
  mutate(sex = case_when(sex == 0 ~ "Male",
                        sex == 1 ~ "Female"),
         sex = fct_relevel(sex, "Male"),
         race2 = case_when(is.na(race) == TRUE ~ "NA",
                          TRUE ~ "non-NA"),
         race = case_when(race == 0 ~ "White",
                          race == 1 ~ "Black",
                          race == 2 ~ "Asian",
                          race == 3 ~ "Hispanic"),
         race = fct_relevel(race, "White"),
         op = as.factor(case_when(op == 0 ~ "Body or tail",
                                  op == 1 ~ "Whipple operation")),
         ven_res = as.factor(case_when(ven_res == 0 ~ "No",
                                       ven_res == 1 ~ "Yes")),
         neo_chemo = as.factor(case_when(neo_chemo == 0 ~ "No",
                                         neo_chemo == 1 ~ "Yes")),
         neo_xrt = as.factor(case_when(neo_xrt == 0 ~ "No",
                                       neo_xrt == 1 ~ "Yes")))

## Rows: 356 Columns: 13
## -- Column specification -----
## Delimiter: ","
## dbl (13): patid, age, sex, race, liver_test, op, ebl, ven_res, neo_chemo, ne...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

#head(pancreatic_df)

```

table 1: demographic characteristics + others a) Grouped by Sex

```

table1 <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
    by = sex,
    type = list(
      ven_res ~ "categorical",
      neo_chemo ~ "categorical",
      neo_xrt ~ "categorical"
    ),
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",      # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)",
      label = list(age ~ "Age (years)",
        race ~ "Race",
        liver_test ~ "Bilirubin Level (in mg/dL)",
        op ~ "Type of Operation",
        ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
        ven_res ~ "Resection + Reconstruction of Major Vasculature",
        neo_chemo ~ "Chemotherapy Prior to Surgery",
        neo_xrt ~ "Radiation Therapy Prior to Surgery",
        les_size ~ "Tumor Size (in cm)",
        or_time ~ "Operation Time (in mins)")) %>%
    add_p() %>%
    add_overall() %>%
    bold_labels() %>%
    modify_caption("***Baseline Characteristics by Sex**")
table1v2 <- table1 %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",   # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()) |>
  gt::gtsave(filename = "figures_tables/table1.pdf")

```

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```
system("pdfcrop figures_tables/table1.pdf figures_tables/table1.pdf")

table1 |>
  as_kable()
```

Table 1: Baseline Characteristics by Sex

Characteristic	Overall N = 356	Male N = 157	Female N = 199	p-value
Age (years)	63 (15)	66 (14)	61 (16)	0.011
Race				<0.001
White	217 (73%)	120 (90%)	97 (59%)	
Asian	9 (3.0%)	5 (3.8%)	4 (2.4%)	
Black	22 (7.4%)	3 (2.3%)	19 (12%)	
Hispanic	49 (16%)	5 (3.8%)	44 (27%)	
Unknown	59	24	35	
Bilirubin Level (in mg/dL)	5.3 (3.6)	5.8 (3.9)	4.8 (3.3)	0.020
Type of Operation				0.035
Body or tail	172 (48%)	66 (42%)	106 (53%)	
Whipple operation	184 (52%)	91 (58%)	93 (47%)	
Estimated Intraoperative Blood Loss (in mL)	658 (92)	675 (89)	644 (93)	0.002
Resection + Reconstruction of Major Vasculature				0.002
No	277 (78%)	110 (70%)	167 (84%)	
Yes	79 (22%)	47 (30%)	32 (16%)	
Chemotherapy Prior to Surgery				0.5
No	289 (81%)	125 (80%)	164 (82%)	
Yes	67 (19%)	32 (20%)	35 (18%)	
Radiation Therapy Prior to Surgery				0.3
No	317 (89%)	137 (87%)	180 (90%)	
Yes	39 (11%)	20 (13%)	19 (9.5%)	
Tumor Size (in cm)	3.28 (2.35)	2.86 (1.71)	3.60 (2.71)	0.043
Operation Time (in mins)	374 (134)	404 (140)	351 (125)	<0.001

b) Grouped by Race

```
table1byrace <- pancreatic_df %>%
  dplyr::select(age:or_time) %>%
  tbl_summary(
    by = race,
    type = list(
      ven_res ~ "categorical",
      neo_chemo ~ "categorical",
      neo_xrt ~ "categorical"
    ),
    statistic = list(all_continuous() ~ "{mean} ({sd})", # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)",
    label = list(age ~ "Age (years)",
      sex ~ "Sex",
      liver_test ~ "Bilirubin Level (in mg/dL)",
```

```

    op ~ "Type of Operation",
    ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
    ven_res ~ "Resection + Reconstruction of Major Vasculature",
    neo_chemo ~ "Chemotherapy Prior to Surgery",
    neo_xrt ~ "Radiation Therapy Prior to Surgery",
    les_size ~ "Tumor Size (in cm)",
    or_time ~ "Operation Time (in mins)")) %>%
add_p() %>%
add_overall() %>%
bold_labels() %>%
modify_caption("**Baseline Characteristics by Race**")

```

59 missing rows in the "race" column have been removed.

```

table1bv2 <- table1byrace %>%
  as_gt() %>%
  gt::tab_options(
    data_row.padding = px(2),
    table.width = pct(100),
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  cols_width(
    label ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()) |>
  gt::gtsave(filename = "figures_tables/table1b.pdf")

```

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```

system("pdftocrop figures_tables/table1b.pdf figures_tables/table1b.pdf")

table1byrace |>
  as_kable()

```

Table 2: Baseline Characteristics by Race

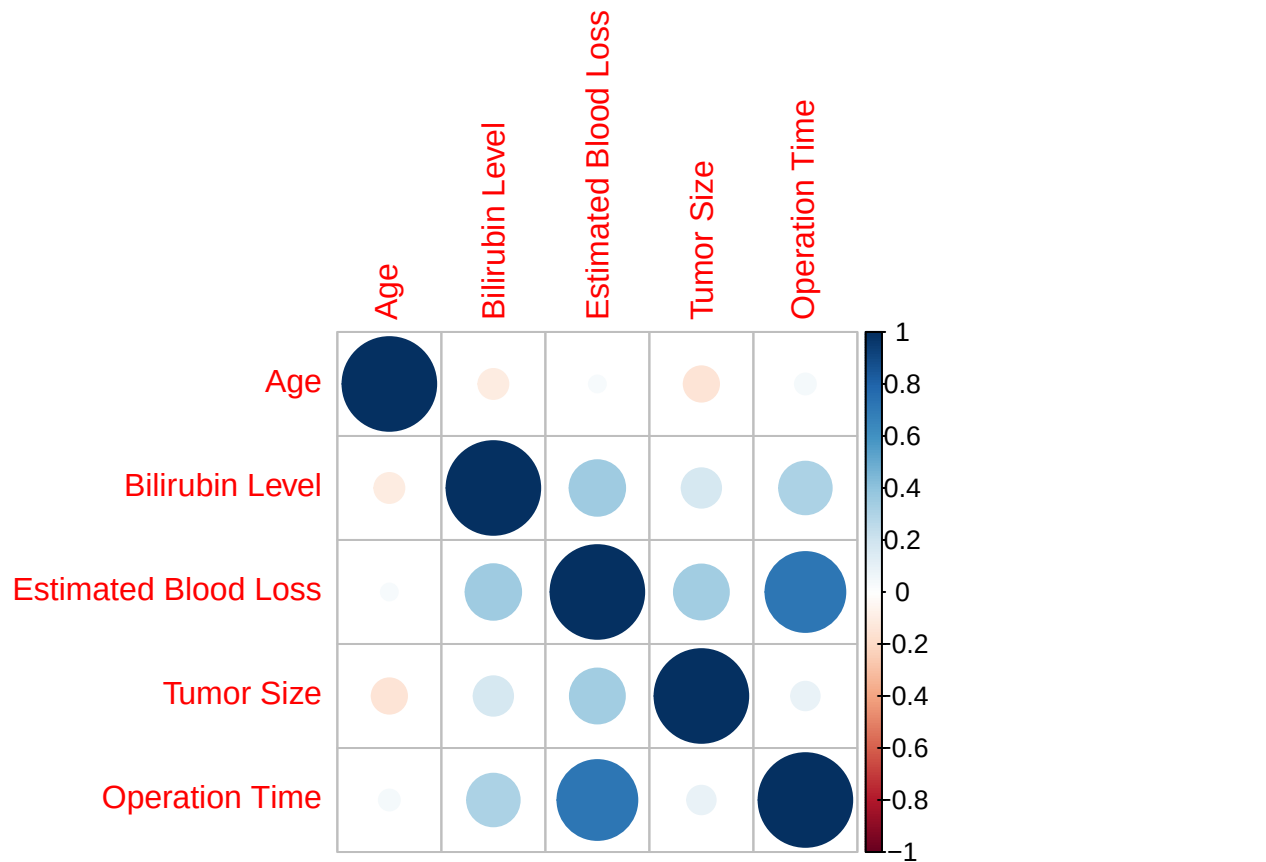
Characteristic	Overall N = 297	White N = 217	Asian N = 9	Black N = 22	Hispanic N = 49	p- value
Age (years)	63 (15)	64 (15)	63 (9)	61 (18)	63 (16)	0.9
Sex						<0.001
Male	133 (45%)	120 (55%)	5 (56%)	3 (14%)	5 (10%)	
Female	164 (55%)	97 (45%)	4 (44%)	19 (86%)	44 (90%)	
Bilirubin Level (in mg/dL)	5.4 (3.7)	5.8 (3.7)	1.9 (2.1)	5.6 (4.8)	4.0 (2.7)	<0.001
Type of Operation						0.4
Body or tail	144 (48%)	109 (50%)	2 (22%)	11 (50%)	22 (45%)	
Whipple operation	153 (52%)	108 (50%)	7 (78%)	11 (50%)	27 (55%)	
Estimated Intraoperative Blood Loss (in mL)	655 (94)	657 (100)	658 (106)	680 (92)	638 (51)	0.2
Resection + Reconstruction of Major Vasculature						<0.001
No	236 (79%)	163 (75%)	7 (78%)	18 (82%)	48 (98%)	
Yes	61 (21%)	54 (25%)	2 (22%)	4 (18%)	1 (2.0%)	
Chemotherapy Prior to Surgery						<0.001
No	240 (81%)	165 (76%)	9 (100%)	18 (82%)	48 (98%)	
Yes	57 (19%)	52 (24%)	0 (0%)	4 (18%)	1 (2.0%)	
Radiation Therapy Prior to Surgery						0.003
No	267 (90%)	187 (86%)	9 (100%)	22 (100%)	49 (100%)	
Yes	30 (10%)	30 (14%)	0 (0%)	0 (0%)	0 (0%)	
Tumor Size (in cm)	3.29 (2.40)	2.98 (1.75)	2.10 (0.74)	5.22 (3.81)	4.01 (3.49)	0.032
Operation Time (in mins)	373 (134)	378 (145)	386 (102)	418 (50)	328 (100)	0.006

###correlation between predictors and within covariates

```
cor_dat <- pancreatic_df |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
pdf("figures_tables/corr_contvars.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
## 2
```

```
corrplot(cor_matrix, method = "circle")
```

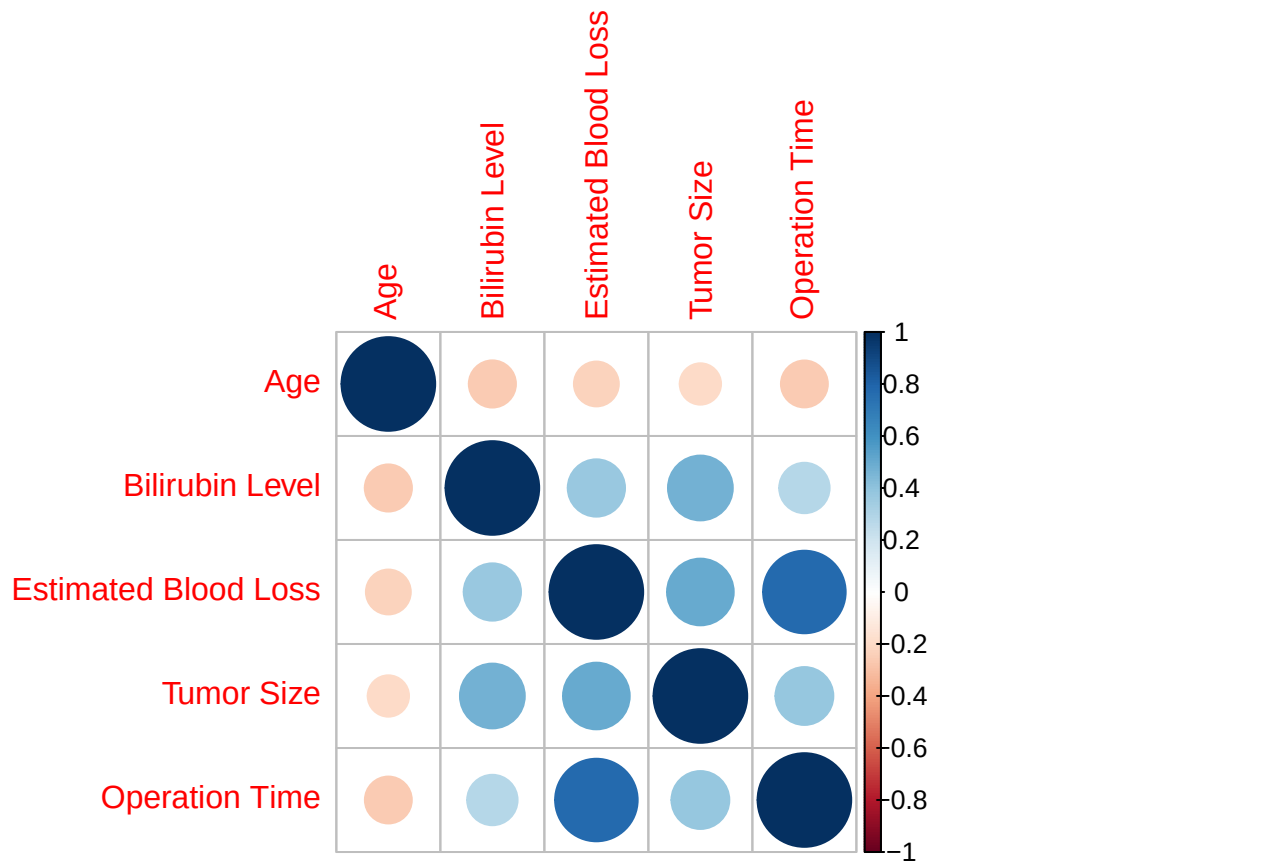


Male

```
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Male") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
pdf("figures_tables/corr_contvars_male.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
## 2
```

```
corrplot(cor_matrix, method = "circle")
```

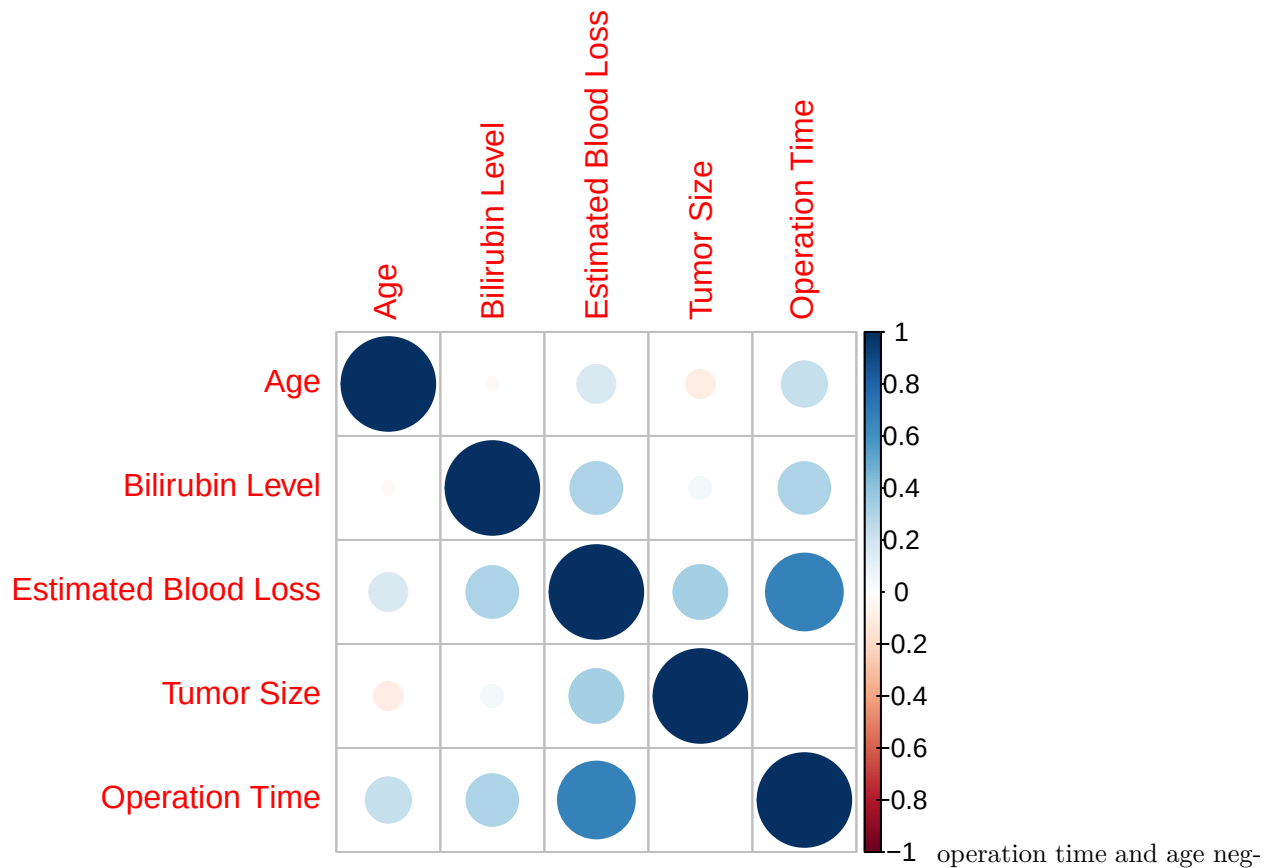


Female

```
cor_dat <- pancreatic_df |>
  dplyr::filter(sex == "Female") |>
  dplyr::select(age, liver_test, ebl, les_size, or_time)
cor_matrix <- cor(cor_dat, method = "pearson")
rownames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
colnames(cor_matrix) <- c("Age", "Bilirubin Level", "Estimated Blood Loss", "Tumor Size", "Operation Time")
pdf("figures_tables/corr_contvars_female.pdf", width = 6, height = 5)
corrplot(cor_matrix, method = "circle")
dev.off()
```

```
## pdf
## 2
```

```
corrplot(cor_matrix, method = "circle")
```



atively correlated in men but no to low positive correlation in women

```
df_filter <- pancreatic_df %>%
  group_by(sex) |>
  mutate(age_quintile = ntile(age, 5)*20) |>
  group_by(sex, age_quintile) |>
  mutate(age_group = paste(min(age), max(age), sep = "-"))

plot <- ggplot(df_filter,
  aes(x = or_time, y = ebl, color = sex, group = sex)) +
  geom_point(size = 0.5, alpha = 0.5) +
  geom_smooth() +
  labs(x = "Operation Time", y = "EBL",
    color = "Sex", title = "Operation Time versus EBL by Age Quintile") +
  facet_wrap(~age_quintile) +
  theme_classic()
pdf("figures_tables/ebl_vs_ortime_age.pdf",
  width = 10, height = 8)
plot
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
dev.off()
```

```
## pdf
## 2
```

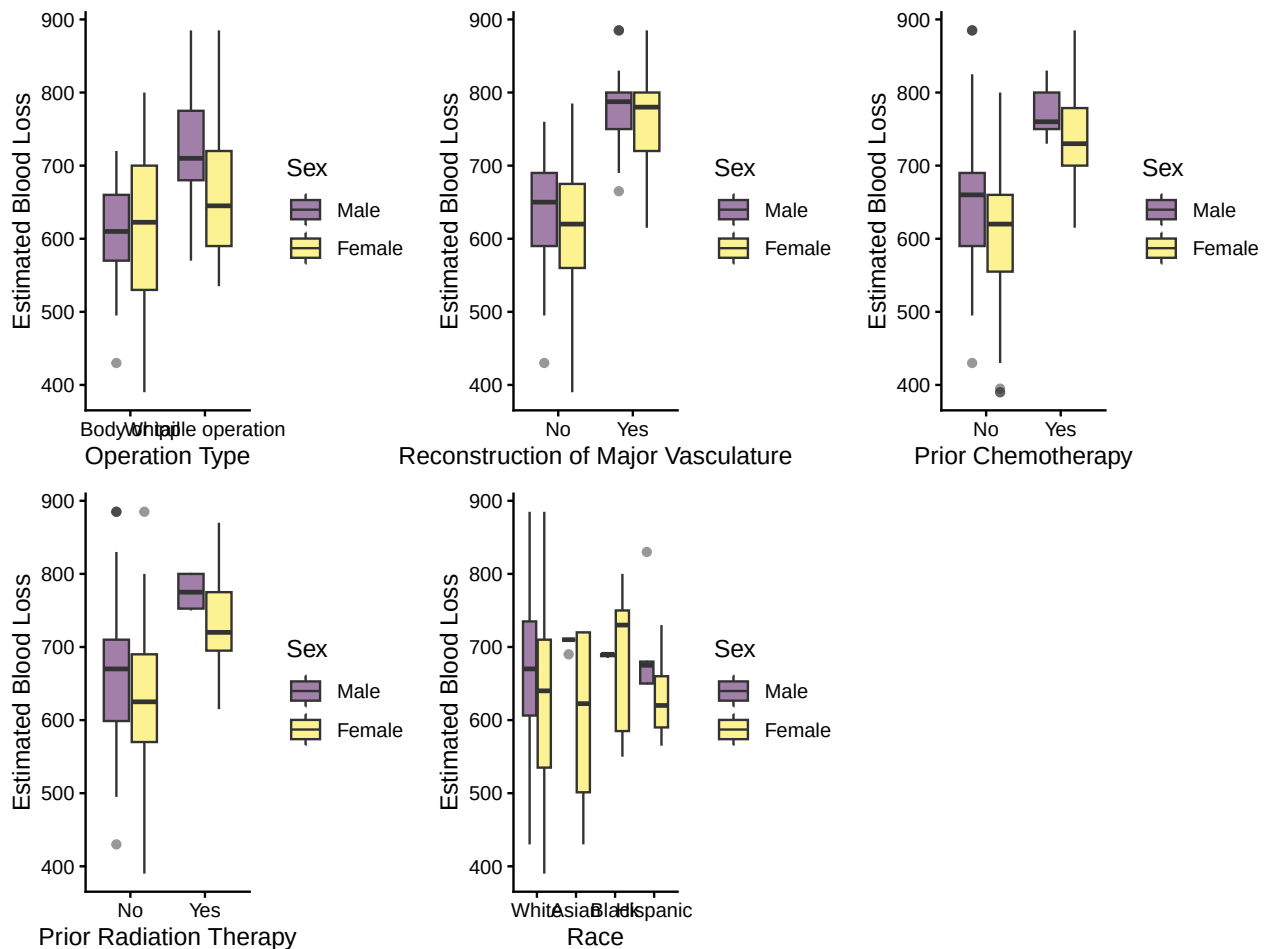

EBL Distribution in each categorical covariates

```

colnames <- c("op", "ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior Radiation Therapy", "Race")
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
    aes(x = get(colnames[x]), y = ebl, fill = sex)) +
    geom_boxplot(alpha = 0.5) +
    labs(x = names[x], y = "Estimated Blood Loss",
      fill = "Sex") +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))

```



```

pdf("figures_tables/ebl_dist_across_cat_boxplot.pdf",
  width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()

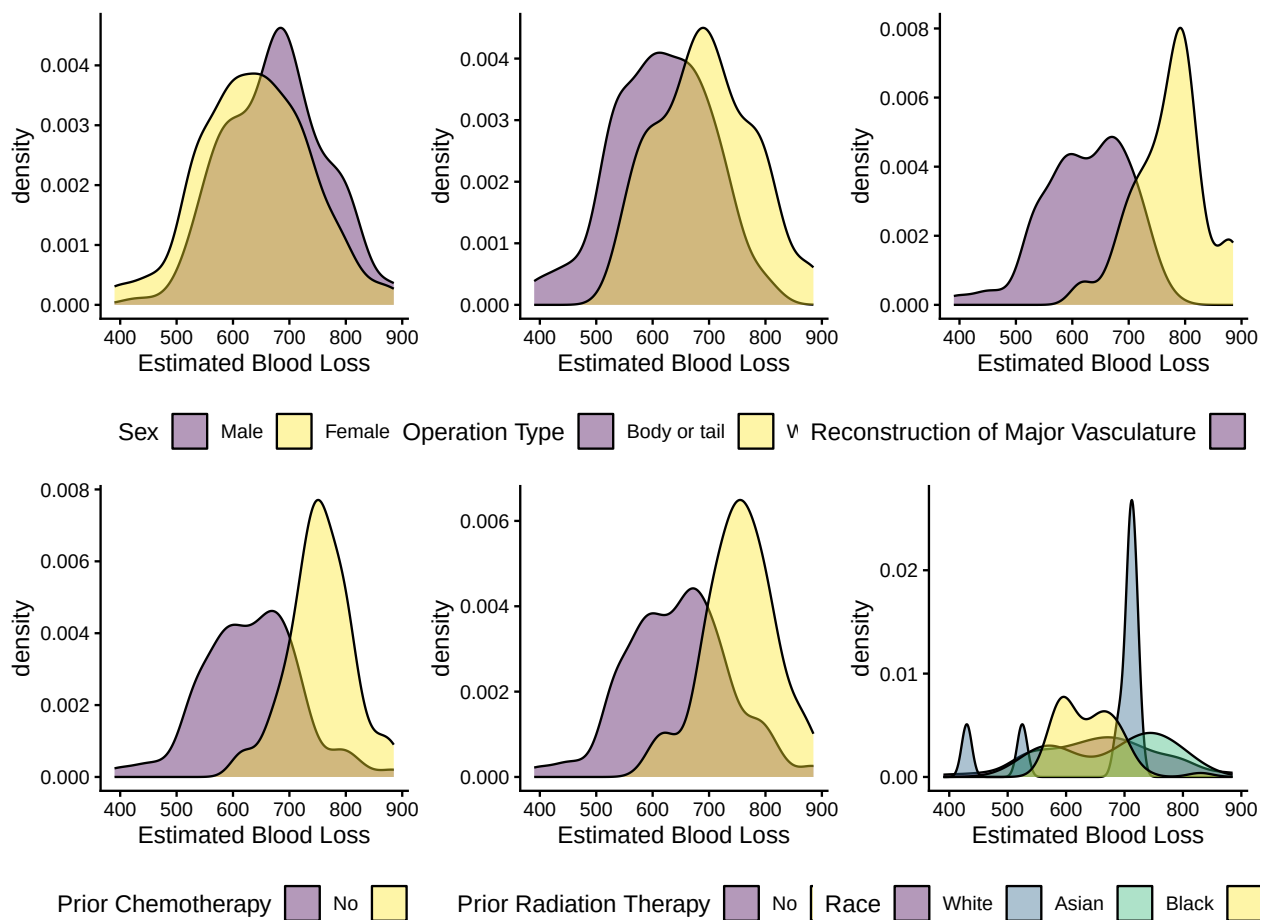
```

```
## pdf
```

```
## 2
```

```
colnames <- c("sex", "op", "ven_res", "neo_chemo", "neo_xrt", "race")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior Radiation Therapy", "Race")
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
    aes(x = ebl, fill = get(colnames[x]))) +
    geom_density(alpha = 0.4) +
    labs(fill = names[x], x = "Estimated Blood Loss") +
    theme_classic() +
    theme(legend.position = "bottom")
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```



```
pdf("figures_tables/ebl_dist_across_cat_densityplot.pdf",
  width = 8, height = 7)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
dev.off()
```

```
## pdf
## 2
```

EBL lower in female regardless of prior chemotherapy, radiation therapy and major reconstruction status. However, EBL in female vs male are different depending on operation type (effect modification probably, can try interaction term). Low EBL in white and hispanic women. Very few asian and black men recruited in the study so comparison is difficult.

EBL's distribution is relatively normal across each group.

Recode race:

```
pancreatic_df <- pancreatic_df |>
  mutate(race = case_when(race == "White" ~ "White",
                          race %in% c("Asian", "Black", "Hispanic") ~ "Non-white"),
         race = fct_relevel(race, "White"))
```

Modeling:

```
lm_v0 <- lm(ebl ~ race*sex + age + op + ven_res + neo_chemo +
            neo_xrt + liver_test + les_size + age*or_time,
            data = pancreatic_df)
```

```
summary(lm_v0)
```

```
##
## Call:
## lm(formula = ebl ~ race * sex + age + op + ven_res + neo_chemo +
##     neo_xrt + liver_test + les_size + age * or_time, data = pancreatic_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -187.174  -25.717   -1.642   35.410  180.361
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)    362.99750    46.11116    7.872 0.0000000000000074 ***
## raceNon-white     31.47342    16.78712    1.875    0.06184 .
## sexFemale       -17.11628     7.91336   -2.163    0.03138 *
## age              1.91624     0.72579    2.640    0.00874 **
## opWhipple operation  7.60825     8.94741    0.850    0.39586
## ven_resYes       28.24232    12.63552    2.235    0.02618 *
## neo_chemoYes     57.32923    12.61133    4.546 0.000008106250132 ***
## neo_xrtYes      -42.19609    15.96244   -2.643    0.00866 **
## liver_test        1.90508     1.00487    1.896    0.05899 .
## les_size         10.99560     1.53084    7.183 0.0000000000006019 ***
## or_time           0.58888     0.12031    4.895 0.000001652846530 ***
## raceNon-white:sexFemale -28.89725    18.70488   -1.545    0.12348
## age:or_time      -0.00437     0.00186   -2.349    0.01951 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 54.78 on 284 degrees of freedom
## (59 observations deleted due to missingness)
## Multiple R-squared:  0.6724, Adjusted R-squared:  0.6586
## F-statistic: 48.59 on 12 and 284 DF,  p-value: < 0.00000000000000022
```

Type of operation not really significant

```
lm_v1 <- lm(ebl ~ race*sex + age + ven_res + neo_chemo +
            neo_xrt + liver_test + les_size + age*or_time,
            data = pancreatic_df)

anova(lm_v1, lm_v0)

## Analysis of Variance Table
##
## Model 1: ebl ~ race * sex + age + ven_res + neo_chemo + neo_xrt + liver_test +
##      les_size + age * or_time
## Model 2: ebl ~ race * sex + age + op + ven_res + neo_chemo + neo_xrt +
##      liver_test + les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      285 854285
## 2      284 852115   1    2169.5 0.7231 0.3959
```

the model with operation time doesn't provide better fit. So removing that

```
lm_v3 <- lm(ebl ~ race*sex + age + ven_res + neo_chemo +
            neo_xrt + liver_test + les_size + age + or_time,
            data = pancreatic_df)
anova(lm_v3, lm_v1)

## Analysis of Variance Table
##
## Model 1: ebl ~ race * sex + age + ven_res + neo_chemo + neo_xrt + liver_test +
##      les_size + age + or_time
## Model 2: ebl ~ race * sex + age + ven_res + neo_chemo + neo_xrt + liver_test +
##      les_size + age * or_time
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      286 871231
## 2      285 854285   1    16946 5.6534 0.01808 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

One with interactions for raceXsex, liver_testXles_size, ageXor_time significantly better

```
#check_model(lm_v1)
#ggsave(filename = "figures_tables/lm_diagnostic_plot.pdf",
#       width = 10, height = 15)
```

Look at missingness

```
table1byrace <- pancreatic_df %>%
  dplyr::select(c(age:or_time, race2)) %>%
  tbl_summary(
    by = race2,
    type = list(
      ven_res ~ "categorical",
      neo_chemo ~ "categorical",
      neo_xrt ~ "categorical"
```

```

),
statistic = list(all_continuous() ~ "{mean} ({sd})",      # drop min/max to save space
  all_categorical() ~ "{n} ({p}%)",
label = list(age ~ "Age (years)",
  sex ~ "Sex",
  liver_test ~ "Bilirubin Level (in mg/dL)",
  op ~ "Type of Operation",
  ebl ~ "Estimated Intraoperative Blood Loss (in mL)",
  ven_res ~ "Resection + Reconstruction of Major Vasculature",
  neo_chemo ~ "Chemotherapy Prior to Surgery",
  neo_xrt ~ "Radiation Therapy Prior to Surgery",
  les_size ~ "Tumor Size (in cm)",
  or_time ~ "Operation Time (in mins)")) %>%
add_p() %>%
add_overall() %>%
bold_labels() %>%
modify_caption("***Baseline Characteristics by Race**")
table1bv2 <- table1byrace %>%
as_gt() %>%
gt::tab_options(
  data_row.padding = px(2),
  table.width = pct(100),
  table.border.top.color = "black",
  table.border.bottom.color = "black",
  heading.border.bottom.color = "black",
  table.font.color = "black",
  column_labels.border.bottom.style = "none", # Remove header line
  table_body.border.bottom.style = "none",    # Remove body line
  table_body.border.top.style = "none",
  table.border.top.style = "none",
  table.font.names = "arial",
  table.font.size = "12px"
) |>
cols_width(
  label ~ px(150),
  everything() ~ px(80)
) |>
gt::tab_style(
  style = gt::cell_borders(),
  locations = gt::cells_body(columns = everything()) |>
gt::gtsave(filename = "figures_tables/table_missingness.pdf")

```

```
## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp60gIG8/fileb45a6cea06.html screenshot
```

```

system("pdftocrop figures_tables/table_missingness.pdf figures_tables/table_missingness.pdf")

table1byrace |>
  as_kable()

```

Table 3: Baseline Characteristics by Race

Characteristic	Overall N = 356	NA N = 59	non-NA N = 297	p-value
Age (years)	63 (15)	62 (16)	63 (15)	0.4
Sex				0.6
Male	157 (44%)	24 (41%)	133 (45%)	
Female	199 (56%)	35 (59%)	164 (55%)	
race				>0.9
White	217 (73%)	0 (NA%)	217 (73%)	
Non-white	80 (27%)	0 (NA%)	80 (27%)	
Unknown	59	59	0	
Bilirubin Level (in mg/dL)	5.3 (3.6)	4.6 (3.4)	5.4 (3.7)	0.14
Type of Operation				0.9
Body or tail	172 (48%)	28 (47%)	144 (48%)	
Whipple operation	184 (52%)	31 (53%)	153 (52%)	
Estimated Intraoperative Blood Loss (in mL)	658 (92)	669 (84)	655 (94)	0.3
Resection + Reconstruction of Major Vasculature				0.092
No	277 (78%)	41 (69%)	236 (79%)	
Yes	79 (22%)	18 (31%)	61 (21%)	
Chemotherapy Prior to Surgery				0.7
No	289 (81%)	49 (83%)	240 (81%)	
Yes	67 (19%)	10 (17%)	57 (19%)	
Radiation Therapy Prior to Surgery				0.2
No	317 (89%)	50 (85%)	267 (90%)	
Yes	39 (11%)	9 (15%)	30 (10%)	
Tumor Size (in cm)	3.28 (2.35)	3.19 (2.13)	3.29 (2.40)	0.7
Operation Time (in mins)	374 (134)	380 (140)	373 (134)	0.8

```
fig2b <- vis_miss(pancreatic_df |> dplyr::select(-prot_energy)) +
  theme(axis.text.x = element_text(vjust = 0))
ggsave("figures_tables/missingness_info.png", fig2b, height = 7, width = 10)
```

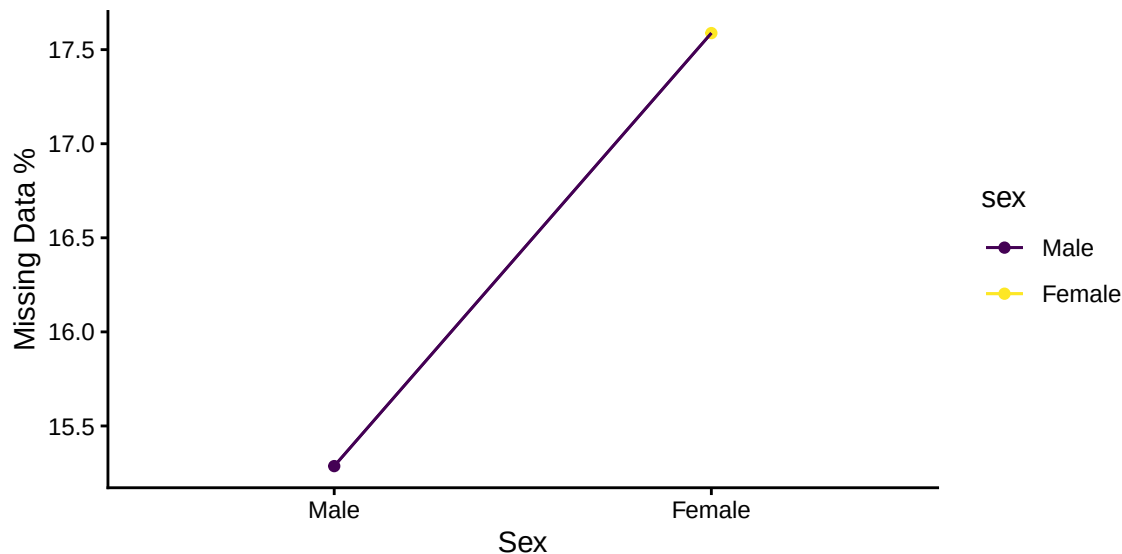
Missingness by gender and age

```
race_gender_missing <- pancreatic_df |>
  dplyr::select(-prot_energy) |>
  group_by(sex) |>
  mutate(total = length(unique(patid))) |>
  ungroup() |>
  drop_na(race) |>
  group_by(sex) |>
  summarize (missing_pct = (total[1] - length(unique(patid)))*100/total[1]) |>
  mutate(dummy_group = 1)

missing_by_gender <- ggplot(aes(x = sex, y = missing_pct, group = dummy_group, color = sex),
  data = race_gender_missing) +
  geom_point() +
  geom_line() +
```

```
geom_line() +
labs(x = "Sex", y = "Missing Data %") +
theme_classic()

ggsave("figures_tables/missingness_byracegender.pdf", missing_by_gender, height = 3, width = 5)
system("pdftocrop figures_tables/missingness_byracegender.pdf figures_tables/missingness_byracegender.pdf")
missing_by_gender
```



Look for other possible sources. - check trend of ebl across each continuous variable to see if its even linear or not, if not linear can try gam

Missingness in different quantile of age or other continuous variables

```
colnames <- c("sex", "op", "ven_res", "neo_chemo", "neo_xrt")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior
categorical_missing_pct <- lapply(1:length(colnames), function(x){
  race_gender_missing <- pancreatic_df |>
    dplyr::select(-prot_energy) |>
    group_by(get(colnames[x])) |>
    mutate(total = length(unique(patid))) |>
    ungroup() |>
    drop_na(race) |>
    group_by(get(colnames[x])) |>
    summarize (missing_pct = (total[1] - length(unique(patid)))*100/total[1]) |>
    mutate(dummy_group = 1)
  colnames(race_gender_missing) <- c(colnames[x], "missing_pct", "dummy_group")
  print(head(race_gender_missing))

  missing_by_cat <- ggplot(aes(x = get(colnames[x]), y = missing_pct, group = dummy_group),
    data = race_gender_missing) +
    geom_point() +
    geom_line() +
    geom_line() +
    labs(x = names[x], y = "Missing Data %") +
    theme_classic()
```

```

return(missing_by_cat)
})

```

```

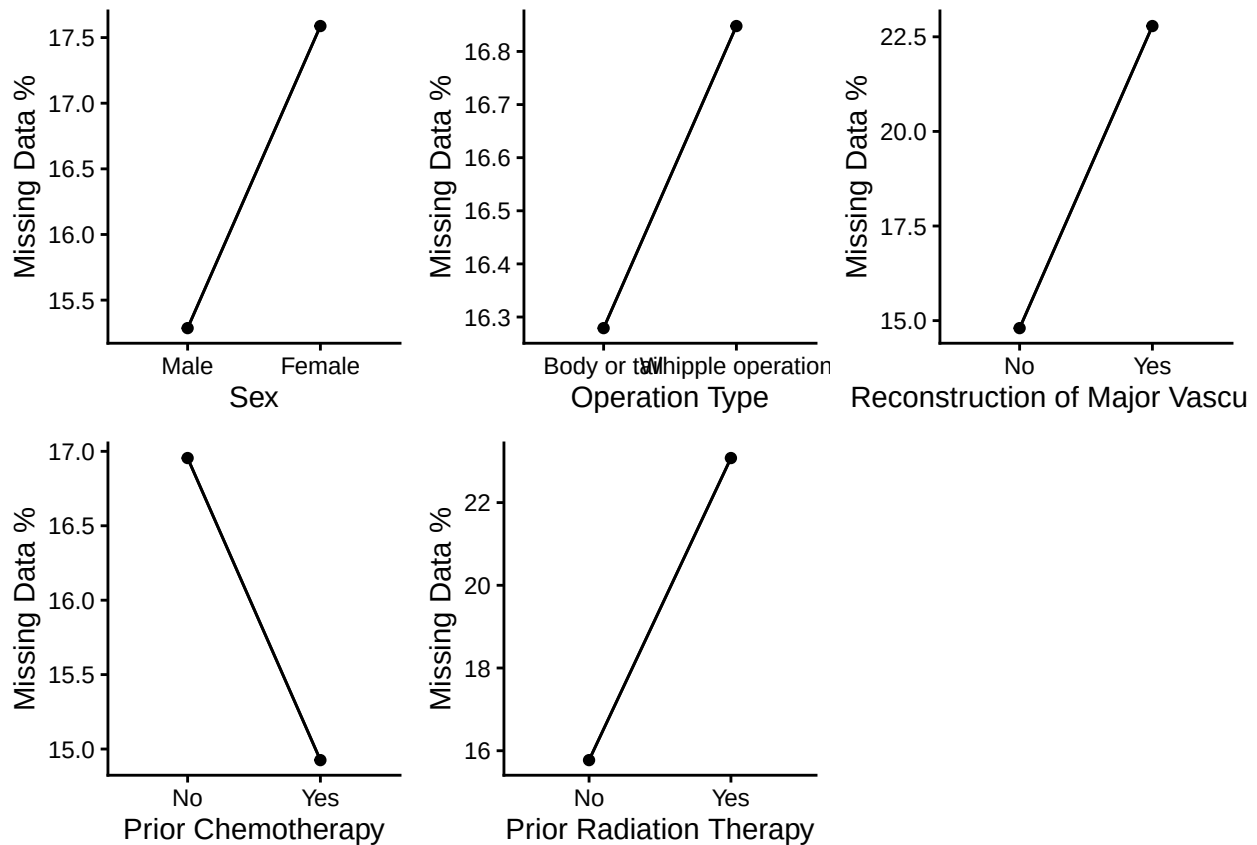
## # A tibble: 2 x 3
##   sex      missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 Male          15.3          1
## 2 Female         17.6          1
## # A tibble: 2 x 3
##   op                missing_pct dummy_group
##   <fct>                <dbl>      <dbl>
## 1 Body or tail          16.3          1
## 2 Whipple operation      16.8          1
## # A tibble: 2 x 3
##   ven_res missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 No          14.8          1
## 2 Yes         22.8          1
## # A tibble: 2 x 3
##   neo_chemo missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 No          17.0          1
## 2 Yes         14.9          1
## # A tibble: 2 x 3
##   neo_xrt missing_pct dummy_group
##   <fct>      <dbl>      <dbl>
## 1 No          15.8          1
## 2 Yes         23.1          1

```

```

do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))

```

```
pdf("figures_tables/missingness_percategorical_group.pdf",
    width = 9, height = 8)
do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))
dev.off()
```

```
## pdf
## 2
```

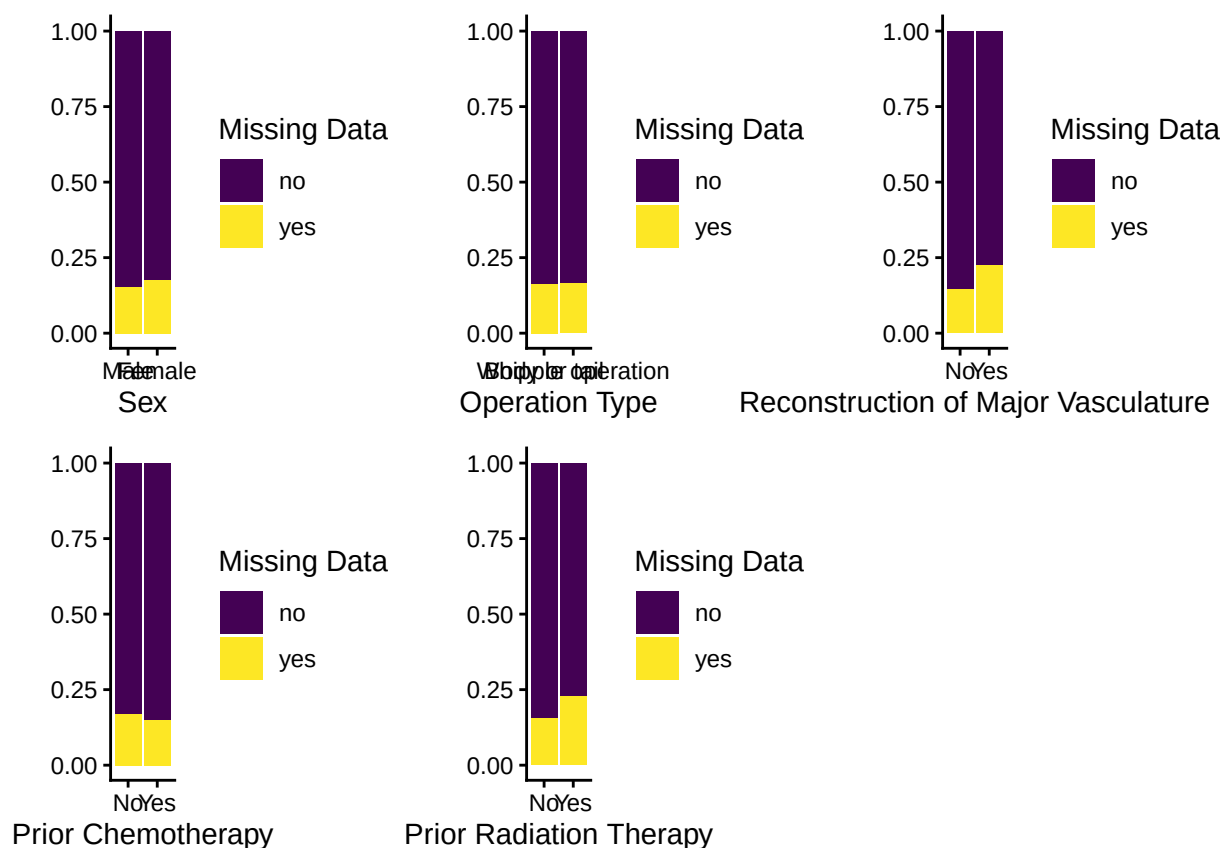
```
colnames <- c("sex", "op", "ven_res", "neo_chemo", "neo_xrt")
names <- c("Sex", "Operation Type", "Reconstruction of Major Vasculature", "Prior Chemotherapy", "Prior
categorical_missing_pct <- lapply(1:length(colnames), function(x){
  a <- pancreatic_df |>
    mutate(xcol = get(colnames[x])) |>
    group_by(xcol, race) |>
    summarise(n = n()) |>
    mutate(sum = sum(n)) |>
    dplyr::filter(is.na(race) == TRUE) |>
    mutate(yes = n/sum, no = 1- yes) |>
    dplyr::select(xcol, n, sum, yes, no) |>
    ungroup() |> pivot_longer(cols = yes:no)

  missing_by_cat <- ggplot(aes(x = xcol, y = value, fill = name),
    data = a) +
    geom_bar(stat = "identity", position = "fill") +
    labs(x = names[x], fill = "Missing Data", y = "") +
    theme_classic()
```

```
return(missing_by_cat)
})
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by xcol and race.
## i Output is grouped by xcol.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(xcol, race))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))
```



```
pdf("figures_tables/missingness_percategorical_group_barplot.pdf",
    width = 11, height = 7)
do.call(gridExtra::grid.arrange, c(categorical_missing_pct, ncol = 3))
dev.off()
```

```
## pdf
## 2
```

```

colnames <- c("ebl", "age", "liver_test", "les_size", "or_time")
names <- c("Estimated Blood Loss", "Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operation")
get_quintile_missing_pct <- lapply(1:length(colnames), function(x){
  df_filter <- pancreatic_df %>%
    group_by(sex) |>
    mutate(xcol = get(colnames[x]),
           x_percentile = ntile(xcol, 20)*5) |>
    group_by(x_percentile, sex) |>
    summarise(n = n(),
              missing_y = sum(is.na(race)),
              group = paste(as.character(min(xcol)), as.character(max(xcol)), sep = "-"),
              covariate = colnames[x],
              pct_missing_y = mean(is.na(race)) * 100)

  plot <- ggplot(df_filter,
                 aes(x = x_percentile, y = pct_missing_y)) +
    geom_point(size = 0.5, alpha = 0.5) +
    labs (x = paste(names[x], "(Percentile)" ), y = "% of Missing Race Info",
          color = "Sex") +
    theme_classic() +
    theme(legend.position = "none")

  return(plot)
})

```

```

## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by x_percentile and sex.
## i Output is grouped by x_percentile.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(by = c(x_percentile, sex))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

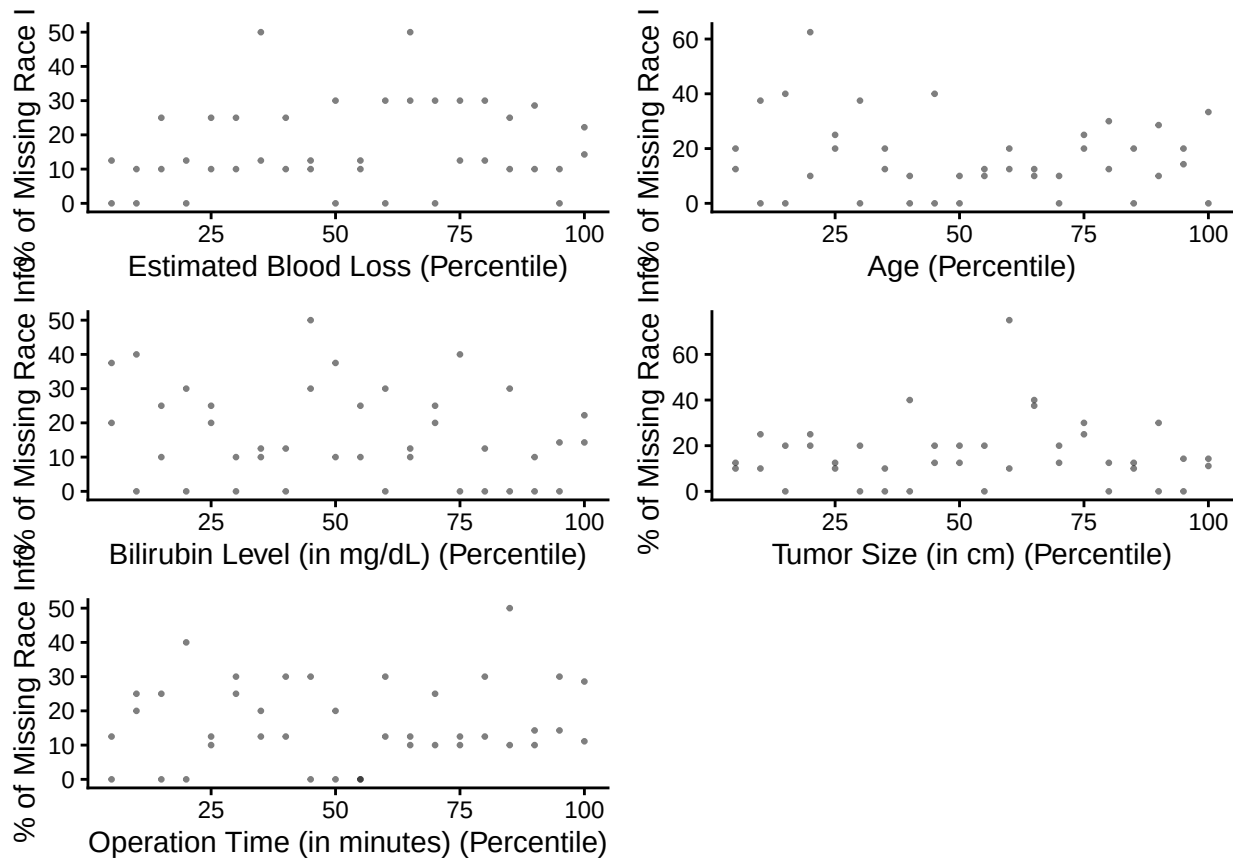
```

```
do.call(gridExtra::grid.arrange, c(get_quintile_missing_pct, ncol = 2))
```

```

## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"

```



```
pdf("figures_tables/missingness_percontinuous_group.pdf",
    width = 10, height = 8)
do.call(gridExtra::grid.arrange, c(get_quintile_missing_pct, ncol = 2))
```

```
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
## Ignoring unknown labels:
## * colour : "Sex"
```

```
dev.off()
```

```
## pdf
## 2
```

The distribution is missing percent against all the continous variables are randomly distributed.

```
mcar_test(pancreatic_df |> dplyr::select(-prot_energy))
```

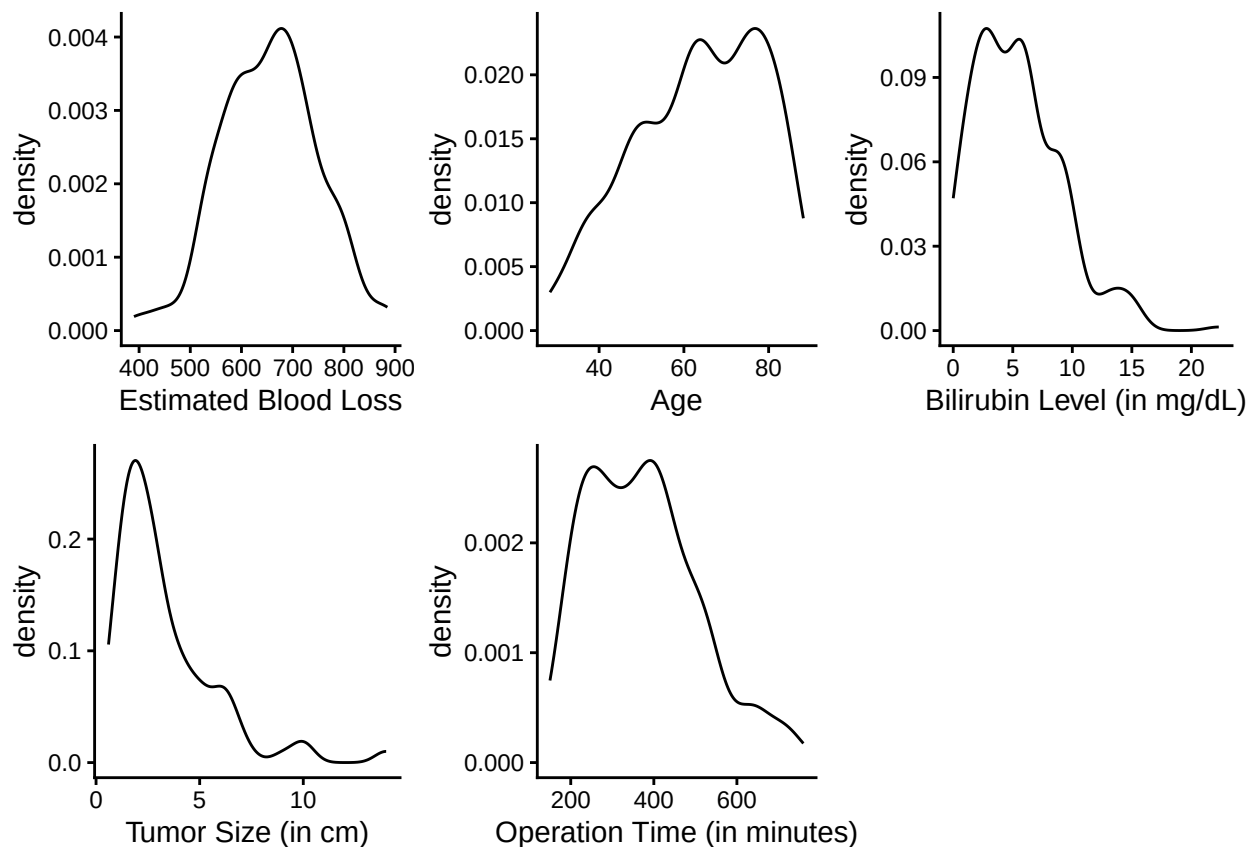
```
## # A tibble: 1 x 4
##   statistic    df p.value missing.patterns
##   <dbl> <dbl>   <dbl>         <int>
## 1     356.    12      0             2
```

significant, reject MCAR, moving forward with MAR assumption

Tumor size (heavily left-skewed)

```
colnames <- c("ebl", "age", "liver_test", "les_size", "or_time")
names <- c("Estimated Blood Loss", "Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operation Time (in minutes)")
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = pancreatic_df |> drop_na(race),
    aes(x = get(colnames[x]))) +
    geom_density(alpha = 0.5) +
    labs(x = names[x]) +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
```



```
pdf("figures_tables/cont_variables_distribution.pdf",
  width = 8, height = 6)
do.call(gridExtra::grid.arrange, c(plots, ncol = 3))
dev.off()
```

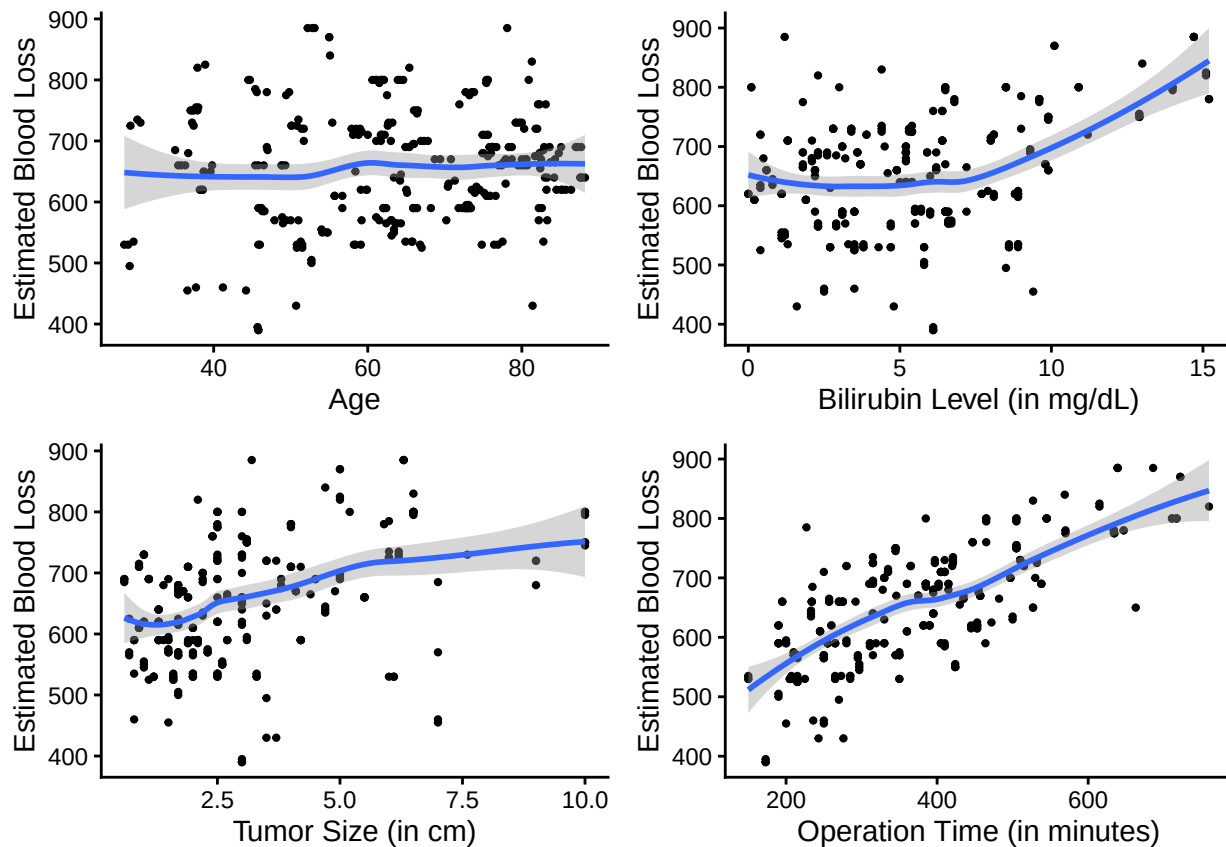
```
## pdf
## 2

outliers_filtered <- pancreatic_df |>
  drop_na(race) |>
  dplyr::filter(les_size < 11,
                liver_test < 20)

colnames <- c("age", "liver_test", "les_size", "or_time")
names <- c("Age", "Bilirubin Level (in mg/dL)", "Tumor Size (in cm)", "Operation Time (in minutes)")
plots <- lapply(1:length(colnames), function(x){
  plot <- ggplot(data = outliers_filtered,
                 aes(x = get(colnames[x]), y = ebl)) +
    geom_point(size = 0.8) +
    geom_smooth() +
    labs(x = names[x], y = "Estimated Blood Loss") +
    theme_classic()
  return(plot)
})

do.call(gridExtra::grid.arrange, c(plots, ncol = 2))

## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



```
pdf("figures_tables/ebl_versus_cont_variable.pdf",
    width = 8, height = 6)
do.call(gridExtra::grid.arrange, c(plots, ncol = 2))
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
dev.off()
```

```
## pdf
## 2
```

Bilirubin level - minimal association at smaller amount, increased association after 5

GAM

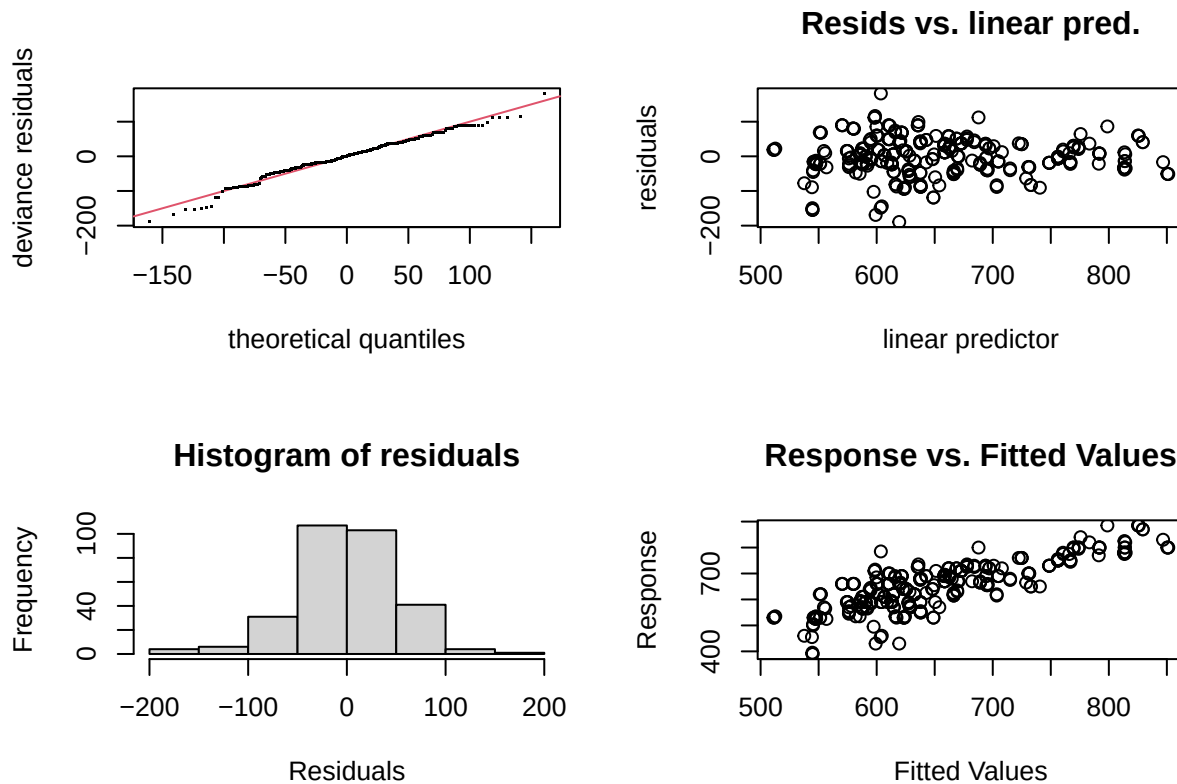
```
gam_v1 <- gam(ebl ~ race*sex + age + sex + ven_res + neo_chemo +
  neo_xrt + s(liver_test) + les_size + age*or_time,
  data = pancreatic_df)

summary(gam_v1)
```

```
##
```

```
## Family: gaussian
## Link function: identity
##
## Formula:
## ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##       s(liver_test) + les_size + age * or_time
##
## Parametric coefficients:
##               Estimate Std. Error t value      Pr(>|t|)
## (Intercept)    366.929266   45.664378    8.035 0.0000000000000249 ***
## raceNon-white    32.897295   16.695284    1.970    0.04975 *
## sexFemale     -16.820837    7.901884   -2.129    0.03414 *
## age             1.993218    0.719776    2.769    0.00599 **
## ven_resYes      29.343916   12.562829    2.336    0.02020 *
## neo_chemoYes     57.381205   12.605056    4.552 0.0000078697847398 ***
## neo_xrtYes     -42.655007   15.945562   -2.675    0.00790 **
## les_size        10.617508    1.464128    7.252 0.0000000000038929 ***
## or_time          0.607858    0.118163    5.144 0.0000005006112286 ***
## raceNon-white:sexFemale -29.041009  18.695029   -1.553    0.12144
## age:or_time      -0.004419    0.001859   -2.378    0.01808 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##               edf Ref.df      F p-value
## s(liver_test)   1      1 3.572 0.0598 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.659   Deviance explained = 67.2%
## GCV = 3123.7   Scale est. = 2997.5      n = 297
```

```
gam.check(gam_v1)
```

```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 14 iterations.
## The RMS GCV score gradient at convergence was 0.0002823353 .
## The Hessian was positive definite.
## Model rank = 20 / 20
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##           k' edf k-index          p-value
## s(liver_test) 9   1   0.66 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
gam_v2 <- gam(ebl ~ race*sex + age + sex + ven_res + neo_chemo +
              neo_xrt + s(liver_test) + les_size + or_time,
              data = pancreatic_df)

summary(gam_v2)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##       s(liver_test) + les_size + or_time
```

```
##
## Parametric coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)   464.31585    19.22051  24.157 < 0.0000000000000002 ***
## raceNon-white    33.11338    16.81643   1.969     0.0499 *
## sexFemale     -19.69836     7.96582  -2.473     0.0140 *
## age            0.45357     0.23628   1.920     0.0559 .
## ven_resYes     23.65153    13.19278   1.793     0.0741 .
## neo_chemoYes    57.98142    12.97133   4.470    0.0000113388757 ***
## neo_xrtYes     -34.29766    15.57159  -2.203     0.0284 *
## les_size       10.21168     1.50639   6.779    0.00000000000706 ***
## or_time         0.33256     0.03716   8.951 < 0.0000000000000002 ***
## raceNon-white:sexFemale -26.02567    18.82211  -1.383     0.1678
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(liver_test) 4.609  5.599 1.793   0.12
##
## R-sq.(adj) =  0.658   Deviance explained = 67.4%
## GCV = 3159.5   Scale est. = 3004.1      n = 297
```

```
anova(gam_v1, gam_v2)
```

```
## Analysis of Deviance Table
##
## Model 1: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##      s(liver_test) + les_size + age * or_time
## Model 2: ebl ~ race * sex + age + sex + ven_res + neo_chemo + neo_xrt +
##      s(liver_test) + les_size + or_time
##   Resid. Df Resid. Dev      Df Deviance      F Pr(>F)
## 1      285.0      854285
## 2      281.4      848326 3.5987   5958.7 0.5512 0.6797
```

```
selected gam_v2
```

```
pdf("figures_tables/gamv1_diagnostic_plot.pdf",
     width = 6, height = 5)
gam.check(gam_v2)
```

```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 4 iterations.
## The RMS GCV score gradient at convergence was 0.002427023 .
## The Hessian was positive definite.
## Model rank = 19 / 19
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##              k'   edf k-index      p-value
```

```
## s(liver_test) 9.00 4.61    0.65 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
dev.off()
```

```
## pdf
##    2
```

```
### Compare fit
```

```
allmods_diag <- data.frame(
  Model = c("LM", "GAM"),
  AIC    = c(AIC(lm_v1), AIC(gam_v2)),
  BIC    = c(BIC(lm_v1), BIC(gam_v2)),
  `log(Likelihood)` = c(as.numeric(logLik(lm_v1)), as.numeric(logLik(gam_v2))),
  DF      = c(attr(logLik(lm_v1), "df"),
               attr(logLik(gam_v2), "df"))
) |>
  kable(caption = "",
        booktabs = FALSE, longtable = FALSE, linesep = "", digits = 2, full_width = FALSE) |>
  kable_styling(latex_options = "hold_position", font_size = 12)

#save_kable(allmods_diag, file = "figures_tables/model_comparison.pdf")
#system("pdftcrop figures_tables/model_comparison.pdf figures_tables/model_comparison.pdf")

allmods_diag
```

Table 4:

Model	AIC	BIC	log.Likelihood.	DF
LM	3234.24	3282.26	-1604.12	13.00
GAM	3237.38	3295.04	-1603.08	15.61

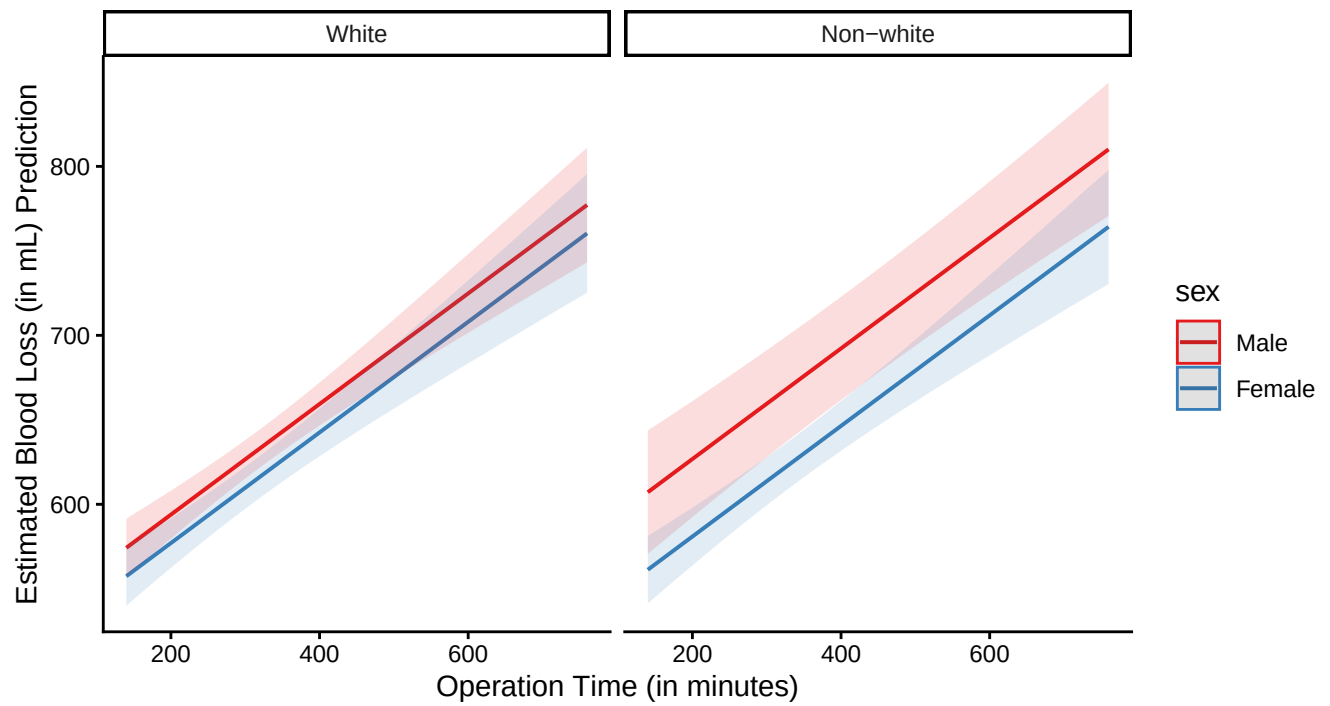
Based on this, `lm_v1` is chosen

```
pdf("figures_tables/gamv1_smoothfunc.pdf", width = 8, height = 8)
plot(lm_v1, shade = TRUE, pages = 1)
dev.off()
```

```
## pdf
##    2
```

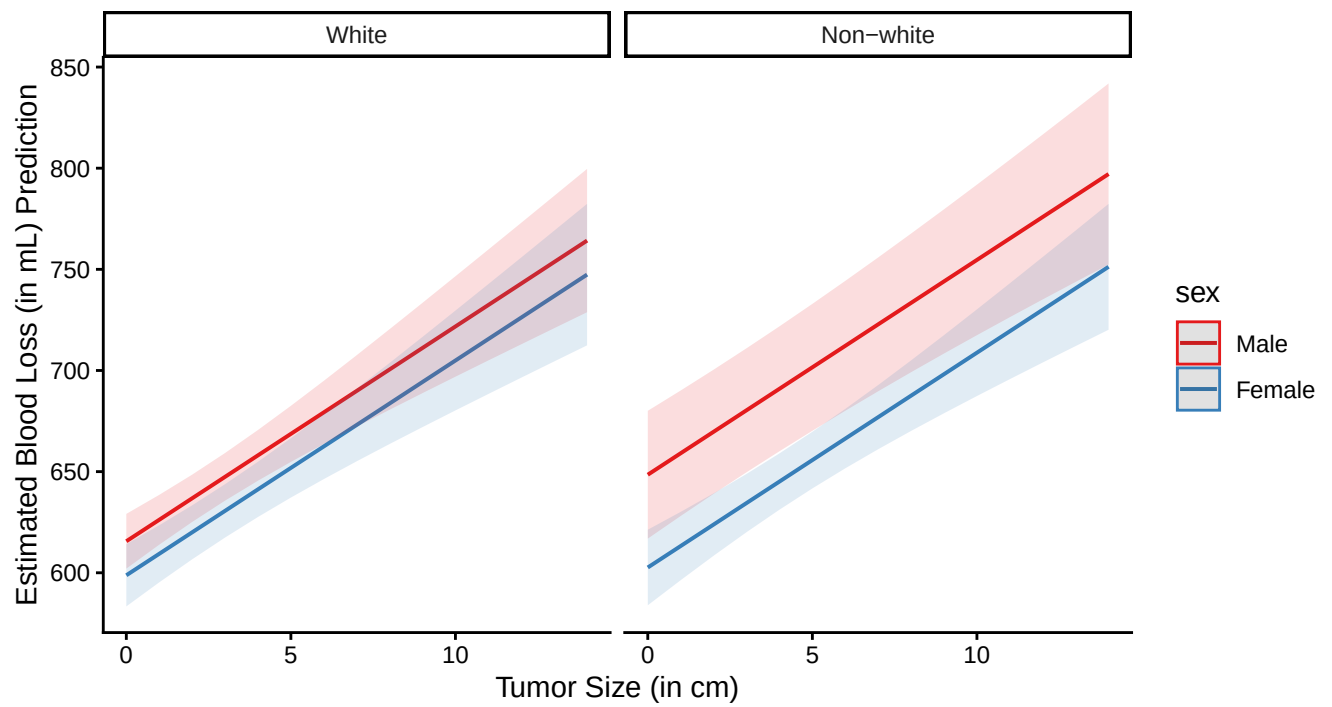
```
#plot(lm_v1, shade = TRUE, pages = 1)
```

```
plot_model(lm_v1, type = "pred", terms = c("or_time", "sex", "race")) +
  labs(title = "",
        y = "Estimated Blood Loss (in mL) Prediction",
        x = "Operation Time (in minutes)") +
  theme_classic()
```



```
ggsave("figures_tables/ebl_pred_race_gender_operatn_time.pdf", height = 4, width = 8)
```

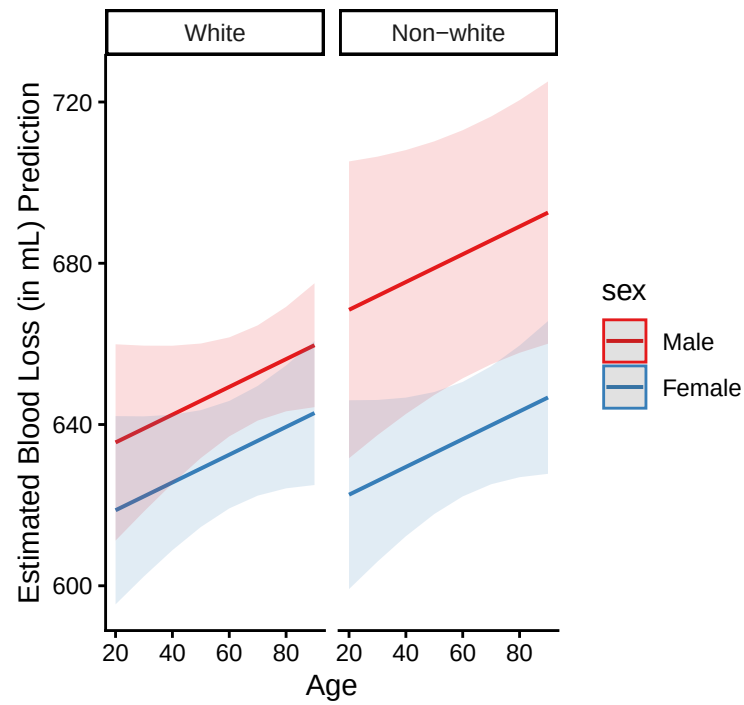
```
plot_model(lm_v1, type = "pred", terms = c("les_size", "sex", "race")) +  
  labs(title = "",  
        y = "Estimated Blood Loss (in mL) Prediction",  
        x = "Tumor Size (in cm)") +  
  theme_classic()
```



```
ggsave("figures_tables/eb1_pred_race_gender_liver_test_tumorsize.pdf", height = 4, width = 8)
```

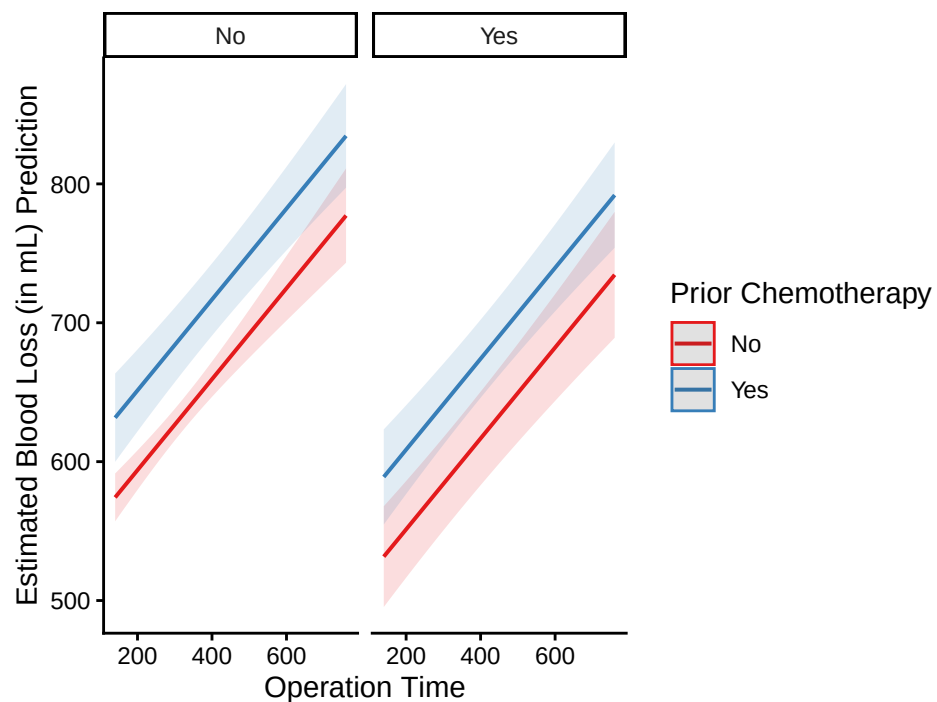
Same trend in men and women across all 4 races

```
plot_model(lm_v1, type = "pred", terms = c("age", "sex", "race"), dodge = 0.6) +
  labs(title = "",
        y = "Estimated Blood Loss (in mL) Prediction",
        x = "Age") +
  theme_classic()
```



```
ggsave("figures_tables/eb1_pred_race_gender_age.pdf", height = 4, width = 8)
```

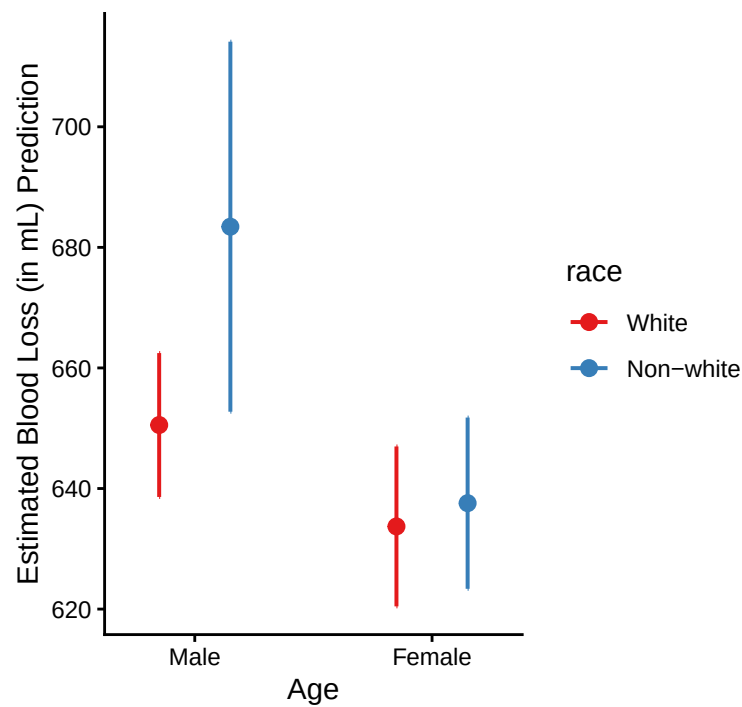
```
plot_model(lm_v1, type = "pred", terms = c("or_time", "neo_chemo", "neo_xrt"), dodge = 0.6) +
  labs(title = "",
        y = "Estimated Blood Loss (in mL) Prediction",
        x = "Operation Time", color = "Prior Chemotherapy") +
  theme_classic()
```



```
ggsave("figures_tables/eb1_pred_chem_rad_therapy.pdf", height = 4, width =8)
```

Consider that normal bilirubin level is 0.2-1.3. Compare the normal state with degree of increase or decrease from there

```
plot_model(lm_v1, type = "pred", terms = c("sex", "race"), dodge = 0.6) +  
  labs(title = "",  
        y = "Estimated Blood Loss (in mL) Prediction",  
        x = "Age") +  
  theme_classic()
```



```
ggsave("figures_tables/eb1_pred_race_gender.pdf", height = 4, width =8)
```

```
##results table
```

```
LM
```

```
bc_tbl <- tbl_regression(
  lm_v1,
  intercept= TRUE,
  conf.int = TRUE,
  p.value = TRUE,
  include = everything(),
  tidy_fun = broom.helpers::tidy_parameters,
  estimate_fun = function(x) sprintf("%.5f", x),
  label = list(age ~ "Age (years)",
               race ~ "Race",
               sex ~ "Sex",
               `liver_test` ~ "Bilirubin Level (in mg/dL)",
               ven_res ~ "Resection + Reconstruction of Major Vasculature",
               neo_chemo ~ "Chemotherapy Prior to Surgery",
               neo_xrt ~ "Radiation Therapy Prior to Surgery",
               `les_size` ~ "Tumor Size (in cm)",
               or_time ~ "Operation Time (in mins)"),
) %>%
  bold_labels() %>%
  modify_header(label ~ "Term")

bc_tbl |>
  as_gt() |>
  gt::tab_options(
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()) |> gt::gtsave("figures_tables/lm_res_tab.pdf")
```

```
## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp60gIG8/fileb45a5d9d6a4b.html screenshot
```

```
system("pdftocrop figures_tables/lm_res_tab.pdf figures_tables/lm_res_tab.pdf")
```

```
bc_tbl |>
  as_kable()
```

Term	Beta	95% CI	p-value
(Intercept)	356.70191	267.16137, 446.24246	<0.001

Term	Beta	95% CI	p-value
Race	NA	NA	
White	NA	NA	
Non-white	32.89730	0.03559, 65.75900	0.050
Sex	NA	NA	
Male	NA	NA	
Female	-16.82084	-32.37429, -1.26738	0.034
Age (years)	1.99322	0.57647, 3.40997	0.006
Resection + Reconstruction of Major Vasculature	NA	NA	
No	NA	NA	
Yes	29.34392	4.61621, 54.07162	0.020
Chemotherapy Prior to Surgery	NA	NA	
No	NA	NA	
Yes	57.38120	32.57039, 82.19202	<0.001
Radiation Therapy Prior to Surgery	NA	NA	
No	NA	NA	
Yes	-42.65501	-74.04102, -11.26900	0.008
Bilirubin Level (in mg/dL)	1.89810	-0.07878, 3.87497	0.060
Tumor Size (in cm)	10.61751	7.73563, 13.49938	<0.001
Operation Time (in mins)	0.60786	0.37527, 0.84044	<0.001
Race * Sex	NA	NA	
Non-white * Female	-29.04101	-65.83886, 7.75684	0.12
Age (years) * Operation Time (in mins)	-0.00442	-0.00808, -0.00076	0.018

GAM

```

bc_tbl <- tbl_regression(
  gam_v2,
  intercept= TRUE,
  conf.int = TRUE,
  p.value = TRUE,
  include = everything(),
  tidy_fun = broom.helpers::tidy_parameters,
  estimate_fun = function(x) sprintf("%.5f", x),
  label = list(age ~ "Age (years)",
               race ~ "Race",
               sex ~ "Sex",
               `s(liver_test)` ~ "s(Bilirubin Level (in mg/dL))",
               ven_res ~ "Resection + Reconstruction of Major Vasculature",
               neo_chemo ~ "Chemotherapy Prior to Surgery",
               neo_xrt ~ "Radiation Therapy Prior to Surgery",
               `les_size` ~ "Tumor Size (in cm)",
               or_time ~ "Operation Time (in mins)"),
  ) %>%
  bold_labels() %>%
  modify_header(label ~ "Term")

```

```

## i Multinomial models, multi-component models and other groups models have a
##   different underlying structure than the models gtsummary was designed for.
## * Functions designed to work with `tbl_regression()` objects may yield
##   unexpected results.
## i Suppress this message with `?suppressMessages()`.

```

```
bc_tbl |>
  as_gt() |>
  gt::tab_options(
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()) |> gt::gtsave("figures_tables/gam_res_tab.pdf")
```

```
## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp60gIG8/fileb45a32e2c4.html screenshot
```

```
system("pdfcrop figures_tables/gam_res_tab.pdf figures_tables/gam_res_tab.pdf")
```

```
bc_tbl |>
  as_kable(digits = 3)
```

Group	Term	Beta	95% CI	p-value
conditional	(Intercept)	464.31585	426.48219, 502.14952	<0.001
_____	Race	NA	NA	
	White	NA	NA	
	Non-white	33.11338	0.01192, 66.21483	0.050
_____	Sex	NA	NA	
	Male	NA	NA	
	Female	-19.69836	-35.37828, -4.01844	0.014
_____	Age (years)	0.45357	-0.01153, 0.91867	0.056
_____	Resection + Reconstruction of Major Vasculature	NA	NA	
	No	NA	NA	
	Yes	23.65153	-2.31713, 49.62020	0.074
_____	Chemotherapy Prior to Surgery	NA	NA	
	No	NA	NA	
	Yes	57.98142	32.44865, 83.51418	<0.001
_____	Radiation Therapy Prior to Surgery	NA	NA	
	No	NA	NA	
	Yes	-34.29766	-64.94877, -3.64655	0.028
_____	Tumor Size (in cm)	10.21168	7.24650, 13.17685	<0.001
_____	Operation Time (in mins)	0.33256	0.25942, 0.40570	<0.001
_____	Race * Sex	NA	NA	
	Non-white * Female	-26.02567	-63.07513, 11.02378	0.2
smooth_terms	s(Bilirubin Level (in mg/dL))	NA	NA	0.12

```
## smooth terms part

smooth_df <- as.data.frame(summary(gam_v2)$s.table)
smooth_df$term <- rownames(smooth_df)

smooth_df <- smooth_df[, c("term", "edf", "F", "p-value")]
colnames(smooth_df) <- c("term", "edf", "statistic", "p.value")

smooth_df <- smooth_df %>% mutate(p.value = case_when(p.value==0~"<0.001", TRUE~as.character(p.value)))
  mutate(term = case_when(term == "s(liver_test)" ~ "s(Bilirubin Level (in mg/dL))"))

smooth_df %>%
  gt() |>
  gt::tab_options(
    table.border.top.color = "black",
    table.border.bottom.color = "black",
    heading.border.bottom.color = "black",
    table.font.color = "black",
    column_labels.border.bottom.style = "none", # Remove header line
    table_body.border.bottom.style = "none",    # Remove body line
    table_body.border.top.style = "none",
    table.border.top.style = "none",
    table.font.names = "arial",
    table.font.size = "12px"
  ) |>
  gt::tab_style(
    style = gt::cell_borders(),
    locations = gt::cells_body(columns = everything()) |>
  tab_header(title = "Smooth Terms") |>
  gt::gtsave("figures_tables/gam_smooth_eff.pdf")
```

```
## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp60gIG8/fileb45a75c05b3b.html screenshot
```

```
system("pdftocrop figures_tables/gam_smooth_eff.pdf figures_tables/gam_smooth_eff.pdf")
```

sensitivity analysis (PMM)

lm_v1 for ebl with all other covariates, excluding race (for NAs and non-NAs)

```
data_complete <- pancreatic_df |>
  dplyr::filter(race2 != "NA")
data_incomplete <- pancreatic_df |>
  dplyr::filter(race2 == "NA")

lm_v1_complete <- lm(ebl ~ sex + age + ven_res + neo_chemo +
  neo_xrt + liver_test + les_size + age*or_time,
  data = data_complete)
lm_v1_incomplete <- lm(ebl ~ sex + age + ven_res + neo_chemo +
  neo_xrt + liver_test + les_size + age*or_time,
  data = data_incomplete)
lm_list <- list(lm_v1_complete,
```

```

      lm_v1_incomplete)
names(lm_list) <- c("LM Complete", "LM Missing Race")

pairs <- combn(names(lm_list), 2, simplify = FALSE)

lapply(pairs, function(compare){
  first_fit <- lm_list[[compare[1]]]
  second_fit <- lm_list[[compare[2]]]
  coef_first <- coef(first_fit)
  coef_second <- coef(second_fit)

  vcov_first <- vcov(first_fit)
  vcov_second <- vcov(second_fit)

  overlapping_terms <- intersect(names(coef_first),
                                names(coef_second))
  res_complete_late <- lapply(overlapping_terms, function(term){

    beta1 <- coef_first[term]
    beta2 <- coef_second[term]

    var1 <- vcov_first[term, term]
    var2 <- vcov_second[term, term]

    z <- (beta1 - beta2) / sqrt(var1 + var2)
    p <- 2 * (1 - pnorm(abs(z)))

    data.frame(
      `Group` = paste0(compare[1], " vs ", compare[2]),
      `Beta1` = beta1, `Beta2` = beta2, `z` = z, `P-value` = p
    )
  })
df = as.data.frame(do.call(rbind, res_complete_late))
nrows = nrow(df)
comp <- knitr::kable(df, digits = 3) |>
  kable_styling(latex_options = c("striped", "hold_position"), font_size = 12) |>
  row_spec(0, extra_css = "border: 1px solid black;") |>
  column_spec(1:6,
    extra_css = "border-left: 1px solid black; border-right: 1px solid black;") |>
  row_spec(1:nrows, extra_css = "border-top: 1px solid black; border-bottom: 1px solid black;")
print(comp)
#save_kable(comp, file = paste0("figures_tables/coef_comparison", compare[1], compare[2], ".pdf"))
})

## \begin{group}\fontsize{12}{14}\selectfont
##
## \begin{longtable}[t]{t}{>{}l>{}l>{}r>{}r>{}r>{}r}
## \toprule
## & Group & Beta1 & Beta2 & z & P.value\\
## \midrule
## \cellcolor{gray!10}{(Intercept)} & \cellcolor{gray!10}{LM Complete vs LM Missing Race} & \cellcolor{gray!10}{sexFemale} & \cellcolor{gray!10}{LM Complete vs LM Missing Race} & -18.696 & 20.707 & -2.316 & 0.021\\
## \cellcolor{gray!10}{age} & \cellcolor{gray!10}{LM Complete vs LM Missing Race} & \cellcolor{gray!10}{

```

```

## ven\_resYes & LM Complete vs LM Missing Race & 26.067 & 18.218 & 0.303 & 0.762\\
## \cellcolor{gray!10}{neo\_chemoYes} & \cellcolor{gray!10}{LM Complete vs LM Missing Race} & \cellcolor{gray!10}{26.067 & 18.218 & 0.303 & 0.762} \\
## \addlinespace
## neo\_xrtYes & LM Complete vs LM Missing Race & -44.636 & -116.274 & 1.080 & 0.280\\
## \cellcolor{gray!10}{liver\_test} & \cellcolor{gray!10}{LM Complete vs LM Missing Race} & \cellcolor{gray!10}{-44.636 & -116.274 & 1.080 & 0.280} \\
## les\_size & LM Complete vs LM Missing Race & 10.800 & 9.526 & 0.364 & 0.716\\
## \cellcolor{gray!10}{or\_time} & \cellcolor{gray!10}{LM Complete vs LM Missing Race} & \cellcolor{gray!10}{10.800 & 9.526 & 0.364 & 0.716} \\
## age:or\_time & LM Complete vs LM Missing Race & -0.005 & -0.001 & -0.833 & 0.405\\
## \bottomrule
## \end{longtable}
## \endgroup{}

## [[1]]
## NULL

```

No significant difference in coefficients so results from CCA is robust.